

↓
Not clear / Unclear / Not exact

The way in which we are handling ambiguity / Not exact
Why we need to use this "Fuzzy Logic"?

In normal logic $\rightarrow$ 1 or 0 / True or False / Yes or No But in Fuzzy logics, things can be in between.

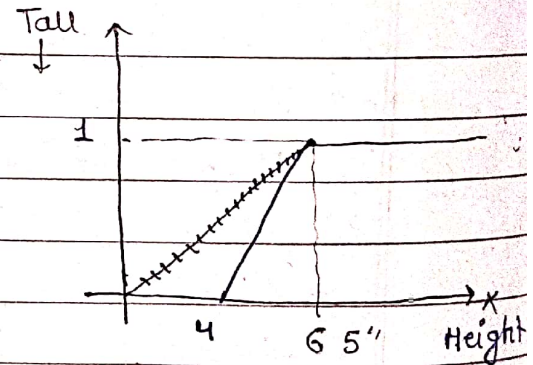
In Fuzzy logic, it is a type of logic in which values are range from 0 to 1, whereas in Normal or crisp logic we have value either 0 or 1 but not in between.

In Normal / Crisp Logic $\rightarrow$ we are defining clear-cut threshold

So fuzzy $\rightarrow$ dealing with maybes, shades, in-between instead of strict yes & no.

Fuzzy Logic $\rightarrow$ Is tolerant of imprecision, uncertainty, partial truth, approximation
 $\rightarrow$ In this logic value can lie anywhere between 0 & 1.
For eg :- In Fuzzy Logic; the concept of "tall" can be considered as :-

Y-axis is representing Tall variable with "membership value".



Y-

Input is height $\rightarrow$ We will tell the O/P in terms of Linguistic Variable
Benefits :- To implement human reasoning in machines

- Apps :
- * Autonomous driving
 - * Washing Machine
 - * Air Conditioning
 - * Cameras

Q. If the loss function is MSE & you have prediction $[2, 0, -1]$ and true value $[-2, 1, -3]$, what is MSE?

$$MSE = \frac{1}{N} \sum_{i=1}^k (y_i - \hat{y}_i)^2 = \frac{1}{3} [(2 - (-2))^2 + (0 - 1)^2 + (-1 - (-3))^2]$$

$$MSE = \frac{1}{3} [4^2 + 1^2 + 2^2] = \frac{1}{3} [16 + 1 + 4] = 7$$

Set → collection of objects having similar characteristics or qualities
 Features: • Crisp boundaries
 • An element either belongs to the set or not

Universe of Discourse (UOD): The set/range of all possible object/values having the same characteristics on a given problem/condition.

For eg: • set of all even numbers between 0 & 100
 • +ve integer numbers $[1, \infty)$ → range is UOD

Universe → X • individual elements in the universe X will be x.
 • Values / feature of the elements in X can be discrete, countable integers or continuous values

• The total no. of elements in a universe X is called Cardinal Number.

* For continuous elements in universe, cardinal no. = ∞ .

UNION:

$$A = \{0, 1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

$$A \cup B = \{0, 1, 2, 3, 4, 5\} \rightarrow \text{"OR Logic" / UNION}$$

Intersection: Elements present in both set. $\equiv$ Logical AND

$$A \cap B = \{3\}$$

The $A \cap B$ includes value x that is in A as well as B

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Page:

Date:

Complement: In the given universe, all the elements apart from the given set is called complement.

$$A^+ = \overset{\text{universe} \rightarrow \text{set of integers}}{\mathbb{Z} - \{0, 1, 2, 3\}}$$

Difference: $A \setminus B = \{x \in A \text{ and } x \notin B\} = A - A \cap B$

$$B - A = B - A \cap B$$

Mid sem

Properties of Classical set:

Commutativity $A \cap B = B \cap A$ $A \cup B = B \cup A$

Associativity $A \cup (B \cap C) = (A \cup B) \cap C$ $A \cap (B \cup C) =$

Distributivity $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Idempotency $A \cup A = A$ $A \cap A = A$

Identify $A \cup \phi = A$

$$A \cap \phi = \phi$$

$$A \cap X = A$$

$$A \cup X = X$$

$X \rightarrow \text{universe}$

Transitivity If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$

Law of excluded middle $A \cup \bar{A} \neq \phi$ } check it

Law of contradiction $A \cap \bar{A} \neq \phi$

De-Morgan Law $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$

$$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

FUZZY SETS: Let $x \rightarrow UOD$ x represents its element s.t. $x \in U$

Then a classical set A can be defined as:

$$A = \{x \mid x \text{ meets certain condition and } x \in X\}$$

$A = \{x \mid 15 < x < 20, x \in X\}$ $X \rightarrow$ all the integers < 20

PRAGATI

Page: _____

Date: _____

$A = \{16, 17, 18, 19\} \rightarrow$ We say membership value of an integer

belonging to set A is one (True) and for all other integers the membership value is zero (False)

Fuzzy theory is based on Multi-valued logic

$A = \{ (16, 0.7), (17, 0.1), (18, 0.2), (19, 0.9) \mid x \in X \}$

$\rightarrow$ 16 belongs to set A with membership value 0.7

Fuzzy Sets: A collection of objects with ismootly boundary

$A = \{ (x, \mu_A(x)) \mid x \in X \}$ $\rightarrow$ Membership value of x

"The membership value of x , present in fuzzy set A is: $\rightarrow x \in X$ " $\rightarrow$ universe

$$\mu_A(x) = \mu_A(16) = 0.7$$

Q. Let $X = \{ \text{Mumbai, Ahmadabad, Surat, IIIT Surat} \}$ be UOD of places where one may choose to live.

Define fuzzy set $A = \text{"Desirable place to live"}$

$A = \{ (\text{Mumbai}, 0.9), (\text{Ahmadabad}, 0.6), (\text{Surat}, 1), (\text{IIIT Surat}, 1) \}$

" $\mu_A(x)$ is membership degree & $\mu_A(x) \in [0, 1]$."

* By default the membership value in classical set is either 0 or 1 not inbetween.

* But in fuzzy set it is in range $[0, 1]$

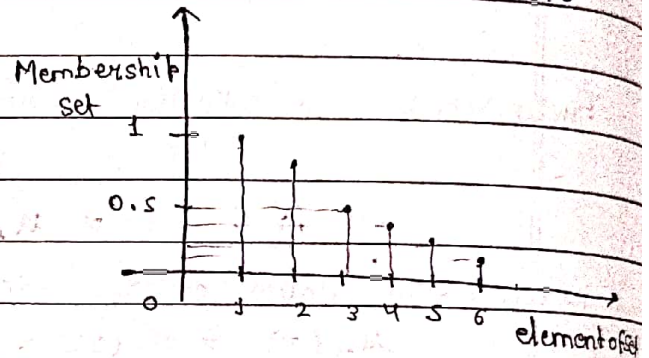
On Discrete Universe of Discourse:-

$$A = 1/1 + 0.8/2 + 0.5/3 + 0.4/4 + 0.3/5 + 0.2/6$$

↗ element of that set

Membership value

$$A = \sum_{x_i \in X} \mu_A(x_i)/x_i$$



Operations on Fuzzy Set :-

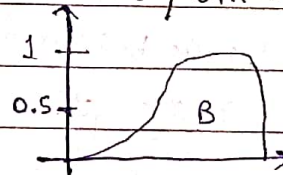
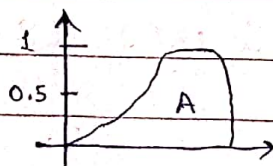
"Visualize"

Equality : Two fuzzy sets of A and B can be considered equal if

$$\mu_A(x) = \mu_B(x), \forall x \in X, y \in Y \text{ and } x = y$$

↗ universe X ↗ universe Y

With same Universe of Discourse / Universe



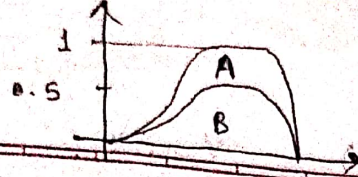
Eg: $A(x) = \{ (x_1, 0.1), (x_2, 0.2), (x_3, 0.3), (x_4, 0.4) \}$

$B(x) = \{ (x_1, 0.1), (x_2, 0.5), (x_3, 0.3), (x_4, 0.6) \}$

As $\mu_A(x) \neq \mu_B(x)$ for $x \in X$, $A(x) \neq B(y)$

Containment (Subset) : A fuzzy set B can be considered as a subset of fuzzy set A if:-

$$\mu_A(x) \geq \mu_B(y), \forall x \in X, y \in Y \text{ and } x = y$$

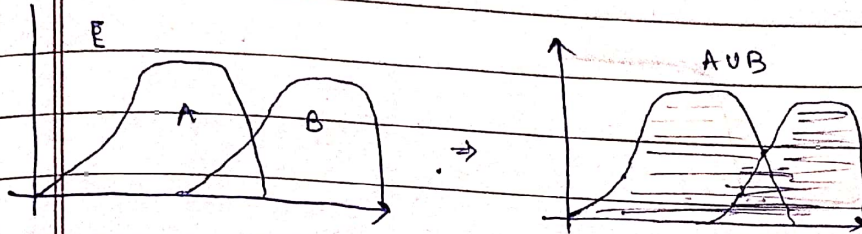


Even though the overlapping part is not present it is true for set B $\subseteq$ set A

Eg: Given eg: $A(x) \subseteq B(x)$

UNION: Union of two fuzzy sets A and B can be defined as:-

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x)) ; \forall x \in X$$



Eg: $A(x) \cup B(x) = B(x)$

Q, $A = 0.1/1 + 0.5/2 + 0.6/4 + 0.3/5$

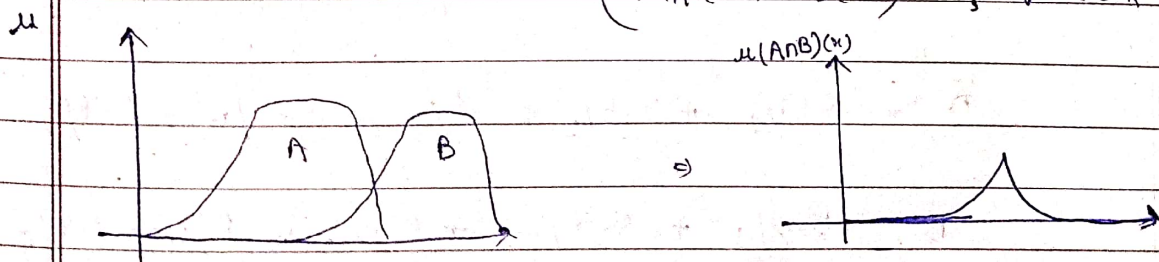
$B = 0.2/1 + 0.3/2 + 0.5/3 + 0.6/5$

Find $A \cup B = 0.2/1 + 0.5/2 + 0.5/3 + 0.6/4 + 0.6/5$

Those elements that are not present are by default having $\mu_A(x) = 0$ or $\mu_B(x) = 0$

INTERSECTION: Intersection of two fuzzy sets A and B can be defined as

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x)) ; \forall x \in X$$

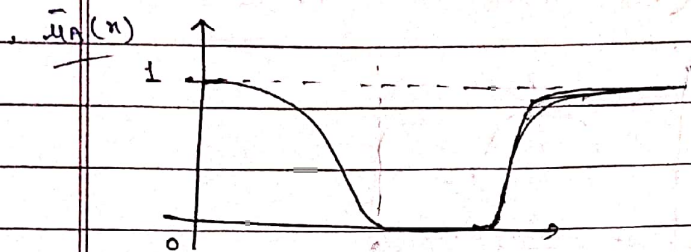
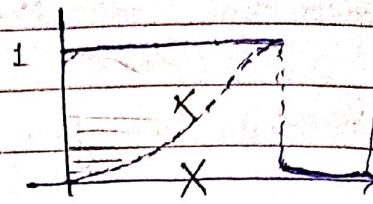
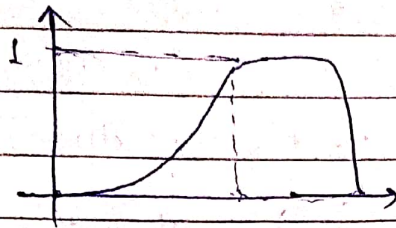


Eg: $A \cap B = 0.1/1 + 0.3/2 + 0.3/5$

If some elements are not mentioned means its membership is 0 and don't need to write

Complement :

$$\bar{\mu}_A(x) = (1 - \mu_A(x)) \quad \forall x \in X$$



$$A = 0.1/1 + 0.5/2 + 0.6/4 + 0.3/5$$

↓
You only have to tell that is not in blue 1 & 5

$$\bar{A} = 1/3 + 0.9/1 + 0.5/2 + 0.4/4 + 0.7/5$$

Q → Proof De-Morgan's Law on A & B

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\Rightarrow \text{LHS: } A \cup B = 0.2/1 + 0.5/2 + 0.5/3 + 0.6/4 + 0.6/5$$

$$\overline{A \cup B} = 0.8/1 + 0.5/2 + 0.5/3 + 0.4/4 + 0.4/5$$

$$\bar{A} = 0.9/1 + 0.5/2 + 1/3 + 0.4/4 + 0.7/5$$

$$\bar{B} = 0.8/1 + 0.7/2 + 0.5/3 + 1/4 + 0.4/5$$

$$\bar{A} \cap \bar{B} = 0.8/1 + 0.5/2 + 0.5/3 + 0.4/4 + 0.4/5$$

Now,

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

PRAGATI

Page: _____

Date: _____

$$A \cap B = 0.1/1 + 0.3/2 + 0.8/3 + 0.3/5$$

$$\bar{A} \cap \bar{B} = 0.9/1 + 0.7/2 + 0.2/3 + 0.7/5 + 1/4$$

$$\bar{A} \cup \bar{B} = 0.9/1 + 0.7/2 + 1/3 + 0.3/5 + 1/4$$

DR. SUDEEP SHARMA

17/09/25

Algebraic product of fuzzy set :-

$$A(x) \cdot B(x) = \{ (x, \mu_A(x) \cdot \mu_B(x)), x \in X \}$$

Eg:

$$A(x) \cdot B(x) = 0.02/1 + 0.15/2 + 0.18/5$$

Multiplying of fuzzy sets by a crisp number : $d \rightarrow$ crisp number

$$A(x) \cdot B(x) = \{ (x, d \cdot \mu_A(x)), x \in X \}$$

* value should not go beyond 1, if product is > 1 consider it 1 only

* if product is giving < 0 , consider it 0.

Power of Fuzzy set: The p^{th} power of a fuzzy set $A(x)$, yields another fuzzy set $A^p(x)$,

$$A^p(x) = \{ (x, \mu_A^p(x)), x \in X \}$$

↓
should be in $[0, 1]$

“Overfitting MODEL $\rightarrow$ DR. SHARMA”

Algebraic sum of two fuzzy set:

$$A(x) + B(x) = \{ (x, \mu_{A+B}(x)), x \in X \}$$

$$\text{where } \mu_{A+B}(x) = \mu_A(x) \overset{\text{plus}}{+} \mu_B(x) - \mu_A(x) \cdot \mu_B(x)$$

Eg:

$$A(x) + B(x) = 0.1 + 0.2 - 0.02/1 + 0.5 + 0.3 - 0.15/2 + 0.6 + 0 - 0/4$$

$$+ 0.5 + 0 - 0/3 + 0.3 + 0.6 - 0.18/5$$

$$A(x) + B(x) = 0.28/1 + 0.65/2 + 0.6/4 + 0.5/3 + 0.72/5$$

Here the membership overall value should not go beyond as it is already managed.

Bounded sum of two fuzzy set:-

$$A(x) \oplus B(x) = \{ (x, \mu_{A \oplus B}(x)), x \in X \}$$

$$\mu_{A \oplus B}(x) = \min \{ 1, \mu_A(x) + \mu_B(x) \}$$

$$A(x) \oplus B(x) = 0.3/1 + 0.8/2 + 0.6/4 + 0.5/3 + 0.9/5$$

Algebraic difference of two fuzzy sets:-

$$A(x) - B(x) = \{ (x, \mu_{A-B}(x)), x \in X \}$$

$$\text{where } \mu_{A-B}(x) = \mu_{A \cap \bar{B}}(x)$$

$$A(x) - B(x) = \{ 0.1/1 + 0.5/2 + 0.6/4 + 0.5/3$$

$$+ 0.3/5$$

$$B(x) - A(x) = \{ (x, \mu_{B-A}(x)), x \in X \}$$

$$\mu_{B-A}(x) = \mu_{\bar{A} \cap B}(x)$$

Bounded difference of two fuzzy sets :

$$A(x) - B(x) = \{ (x, \mu_{A \ominus B}(x)), x \in X \}$$

$$\mu_{A \ominus B}(x) = \max\{0, \mu_A(x) + \mu_B(x) - 1\}$$

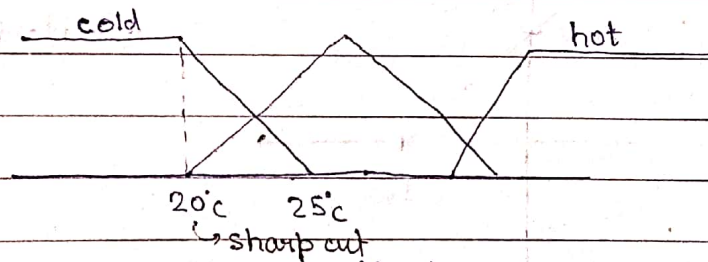
$$A(x) - B(x) =$$

Cartesian product of two fuzzy sets : $A(x) \rightarrow$ universal set X
 $B(y) \rightarrow$ universal set Y

$$\mu_{A \times B}(x, y) = \min\{ \mu_A(x), \mu_B(y) \} \quad x \in X, y \in Y$$

FUZZY RELATION :

A fuzzy relation represents presence or absence of association, interaction, or interconnection b/w elements of 2 or more sets.



The concept can be generalised to various degrees or strengths of association or interaction b/w elements

Crisp Relation :

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 5, 7\}$$

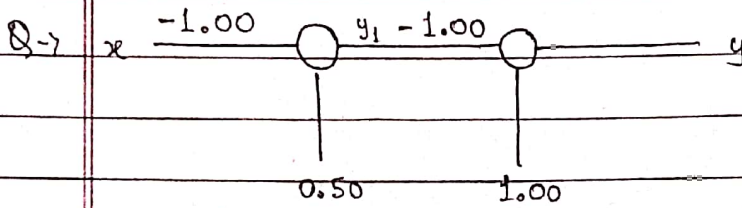
Relation R is defined on $A \times B$

$$A \times B = \{(1, 3), (1, 5), (1, 7), (2, 3), (2, 5), (2, 7), (3, 3), (3, 5), (3, 7), (4, 3), (4, 5), (4, 7)\}$$

$$R = \{(x, y) : y = x + 1, (x, y) \in A \times B\}$$

$$R = \{(1, 2), (2, 3)\}$$

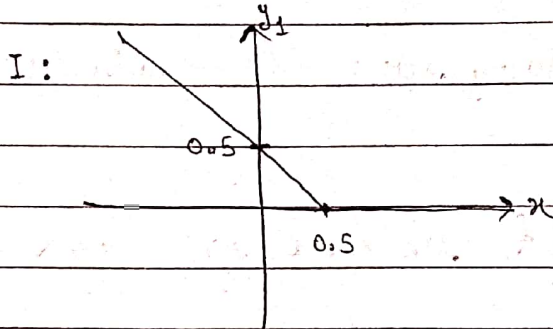
MID SEM QUESTIONS



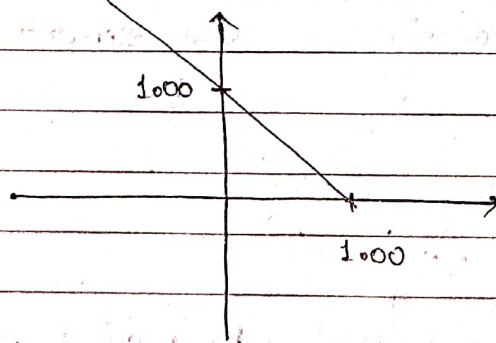
Assuming Neurons have ReLU activation f^*

For $-2 < x < 2$

$$y_1 = \max(-x + 0.5, 0)$$

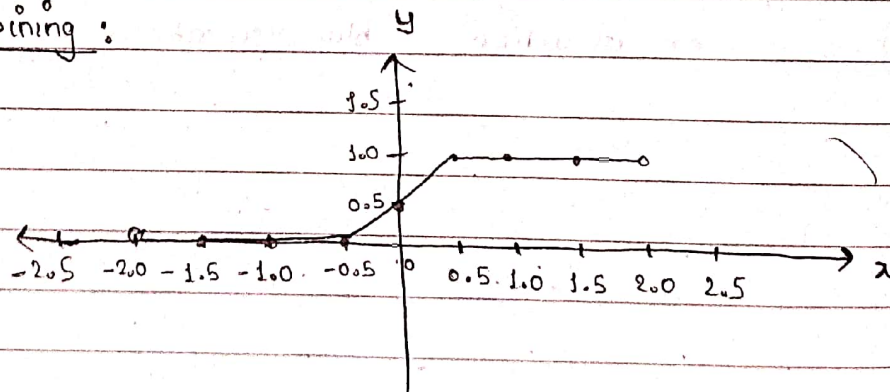


II: $y = -1.00 y_1 + 1.00 \equiv -x + 1.00$



$$y = \max(0, y_1)$$

Combining :



Take $x = 0.5$, $y_1 = 0$ $y = 1.0$

Take $x = 1.0$ $y_1 = 0$ $y = 1.0$

Take $x = 1.5$ $y_1 = 0$ $y = 1.0$

Take $x = 0$ $y_1 = 0.5$ $y = 0.5$

Take $x = -0.5$ $y_1 = 1.0$ $y = 0.0$

Take $x = -1.0$ $y_1 = 1.5$ $y = 0.0$

Take $x = -1.5$ $y_1 = 2.0$ $y = 0.0$

PRAGATI

Page

Date

Q.

$$MSE = \frac{1}{2N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$w_1 = 1$$

$$x_1 = 1$$

$$w_2 = 1$$

$$x_2 = 1$$

$$b = 1$$

$$\text{Target } y = 5$$

ReLU activation F' $\eta = 0.5$

$$y = \max(x, 0)$$

$$\text{Using Backpropagation Algo: } \frac{dy}{dx_1} = x_1, \frac{dy}{dw_2} = x_2, \frac{dy}{db} = 1$$

Iteration 1 :

Forward pass

$$\hat{y}_1 = \max(x, 0) = \max(3, 0) = 3$$

$$x = 1 \times 1 + 1 \times 1 + 1 = 3$$

Loss calculation

$$MSE = \frac{1}{2N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$MSE = \frac{1}{2 \times 1} (5 - 3)^2 = \frac{1}{2} \times 2 \times 2 = 2$$

Backward pass :

$$\frac{dLoss}{dw_1} = \frac{-1}{2N} \times 2 (y_i - \hat{y}_i) \cdot x_1 = \frac{1}{1} \times (5 - 3) \cdot 1 = 2$$

$$\frac{dLoss}{dw_2} = \frac{-1}{2N} \times 2 (y_i - \hat{y}_i) \cdot x_2 = \frac{1}{1} \times (5 - 3) \cdot 1 = 2$$

$$\frac{dLoss}{db} = \frac{-1}{2N} \times 2 (y_i - \hat{y}_i) \cdot 1 = \frac{1}{1} \times (5 - 3) = 2$$

weight update

$$w_1(\text{new}) = w_1(\text{old}) - \eta \frac{d\text{Loss}}{dw_1} = 1 + \frac{1 \times 2}{2} = 2$$

$$w_2(\text{new}) = w_2(\text{old}) - \eta \frac{d\text{Loss}}{dw_2} = 1 + \frac{1 \times 2}{2} = 2$$

$$b(\text{new}) = b(\text{old}) - \eta \frac{d\text{Loss}}{db} = 1 + \frac{1 \times 2}{2} = 2$$

After iteration 1

$$w_1 = 2 \quad w_2 = 2 \quad b = 2$$

$$x_1 = 1 \quad x_2 = 1 \quad y = \text{Target} = 5$$

Iteration 2

Forward pass : $\hat{y}_i = \max(z, 0)$

$$z = 2 \times 1 + 2 \times 1 + 2 = 6$$

$$y_i = 6$$

Loss calculation : $\text{MSE} = \frac{1}{2 \times 2} [(5-3)^2 + (5-6)^2]$

$$\text{MSE} = \frac{1}{2} \times (5-6)^2 = \frac{1}{2} = 0.5$$

Backward pass :

$$\frac{\partial \text{Loss}}{\partial w_1} = \frac{1}{2 \times N} \times 2 (y_i - \hat{y}_i) \times x_1 = \frac{1}{1} \times (5-6) \times 1 = -1$$

$$\frac{\partial \text{Loss}}{\partial w_2} = \frac{1}{2 \times N} \times 2 (y_i - \hat{y}_i) \times x_2 = \frac{1}{1} \times (5-6) \times 1 = -1$$

$$\frac{\partial \text{Loss}}{\partial b} = \frac{1}{2 \times N} \times 2 (y_i - \hat{y}_i) \times 1 = \frac{1}{1} \times (5-6) \times 1 = -1$$

Weight update Rule :

PRAGATI

Page: _____

Date: _____

$$w_1(\text{new}) = 0 - \frac{1}{2} \times \frac{5}{2} = -5/4 = 2 - \frac{1}{2} \times 1 = 3/2$$

$$w_2(\text{new}) = 0 + \frac{1}{2} \times \frac{5}{2} = 5/4 = 2 - \frac{1}{2} \times 1 = 3/2$$

$$b(\text{new}) = 0 - \frac{1}{2} \times \frac{5}{2} = -5/4 = 2 - \frac{1}{2} \times 1 = 3/2$$

After iteration 2 :

$$x_1 = 1$$

$$w_1 = 1.5$$

$$b = 1.5$$

$$x_2 = 1$$

$$w_2 = 1.5$$

End Sem 1. More question

Draw graph using Neural Network.

The Loss value f'' will decrease not the confident about weights & bias.

Fuzzy Relation

What is the utility of Relation? $\rightarrow$ To know the condition or criteria that two or more entities may bear.

Relationship is first applied in Cartesian product of sets

Cartesian product of $A = \{1, 2, 3, 4\}$ $B = \{3, 5, 7\}$

$$A \times B = \begin{matrix} & \begin{matrix} 3 & 5 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

$$R = \{ (x, y) \mid y = x + 1, (x, y) \in A \times B \}$$

$$\begin{matrix} & \begin{matrix} 3 & 5 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$1 \rightarrow$ denoting the relation
Satisfied for given
Combination

Fuzzy Relation:- It is defined on the Cartesian product of sets $X_1, X_2, \dots, X_n$

* Here n -tuples $(x_1, x_2, \dots, x_n)$ may have varying degree of membership within the relationship.

* Membership values $\rightarrow$ indicates the strength of relation b/w the tuples

Operation on Fuzzy Relation:-

Let R and S be two fuzzy relation on $A \times B$

* Union

* Intersection

*

* Composition :

$$T = R \circ S$$

PRAGATI

Page: _____

Date: _____

$$\mu_{ROS}(x, z) = \max_{y \in Y} \{ \min(\mu_R(x, y), \mu_S(y, z)) \}$$

"It is also known as Max-min composition"

Eg:

$$X = (x_1, x_2, x_3)$$

$$Y = (y_1, y_2)$$

$$Z = (z_1, z_2, z_3)$$

$$R = \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.5 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \end{matrix} \quad 3 \times 2$$

$$S = \begin{matrix} & z_1 & z_2 & z_3 \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.6 & 0.4 & 0.7 \\ 0.5 & 0.8 & 0.9 \end{bmatrix} \end{matrix} \quad 2 \times 3$$

1 question End Sem

$$\mu_{ROS}(x, z) = \max_{y \in Y} \{ \min(\mu_R(x, y), \mu_S(y, z)) \}$$

Do not try to solve this in 1 step!!

	z_1	z_2	z_3
x_1	0.5	0.4	0.5
x_2	0.6	0.8	0.9
x_3	0.6	0.6	0.7

3x3

Very high possibility to do Mistake!!

$$\begin{aligned} \mu_{ROS}(x_1, z_1) &= \max \{ \min(\mu_R(x_1, y_1), \mu_S(y_1, z_1)), \\ &\quad \min(\mu_R(x_1, y_2), \mu_S(y_2, z_1)) \} \\ &= \max \{ \min(0.5, 0.6), \min(0.1, 0.5) \} \\ &= \max \{ 0.5, 0.1 \} \\ &= 0.5 \end{aligned}$$

$$\mu_{ROS}(x_1, z_2) = \max \{ \min(\mu_R(x_1, y_1), \mu_S(y_1, z_2)), \min(\mu_R(x_1, y_2), \mu_S(y_2, z_2)) \}$$

$$\mu_{ROS}(x_1, z_3) = \max \{ \min(\mu_R(x_1, y_1), \mu_S(y_1, z_3)), \min(\mu_R(x_1, y_2), \mu_S(y_2, z_3)) \}$$

$$\mu_{ROS}(x_2, z_1) = \max \{ \min(\mu_R(x_2, y_1), \mu_S(y_1, z_1)), \min(\mu_R(x_2, y_2), \mu_S(y_2, z_1)) \}$$

$$\mu_{ROS}(x_2, z_2) = \max \{ \min(\mu_R(x_2, y_1), \mu_S(y_1, z_2)), \min(\mu_R(x_2, y_2), \mu_S(y_2, z_2)) \}$$

COMPOSITION

$$T = R \circ S$$

$$\mu_{ROS}(x, y) = \max_{y \in Y} \{ \mu_R(x, y) \cdot \mu_S(y, z) \}$$

It is also known as Max-dot or max-product composition

$$T_{\text{max-dot}} = R \circ S$$

	z_1	z_2	z_3
x_1	0.3	0.2	0.35
x_2	0.45	0.72	0.81
x_3	0.48	0.48	0.56

$$\mu_{ROS}(x_1, z_1) = \max \{ \mu_R(x_1, y_1) \cdot \mu_S(y_1, z_1), \mu_R(x_1, y_2) \cdot \mu_S(y_2, z_1) \}$$

$$\mu_{ROS}(x_1, z_2) = \max \{ \mu_R(x_1, y_1) \cdot \mu_S(y_1, z_2), \mu_R(x_1, y_2) \cdot \mu_S(y_2, z_2) \}$$

$$\mu_{ROS}(x_1, z_3) = \max \{ \mu_R(x_1, y_1) \cdot \mu_S(y_1, z_3), \mu_R(x_1, y_2) \cdot \mu_S(y_2, z_3) \}$$

Advantage:- When He occurs in Max-min, then it can be solved using max-product.

Fuzzy Tolerance & Equivalence Relation :

A fuzzy relⁿ, R on a single UoD X is also a relation from X to X. It is a fuzzy ^{equivalence} relation if all three of the following property for matrix relⁿ define it:-

Reflexivity $\mu_R(x_i, x_i) = 1 \rightarrow$ ^{Diagonal element of Matrix} should be 1 only.

Symmetry $\mu_R(x_i, x_j) = \mu_R(x_j, x_i)$

Transitivity $\mu_R(x_i, x_j) = \lambda_1$ & $\mu_R(x_j, x_k) = \lambda_2$

$\rightarrow \mu_R(x_i, x_k) = \lambda$ where $\lambda \geq \min[\lambda_1, \lambda_2]$

Tolerance Relⁿ : This relation satisfies Reflexivity, Symmetry, BUT NOT necessarily transitivity.

Page: _____
Date: _____

Reflexivity $\mu_R(x_i, x_i) = 1$

Q →

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 1 & 0.8 & 0.6 & 0.2 \\ 0.8 & 1 & 0.7 & 0.3 \\ 0.6 & 0.7 & 1 & 0.5 \\ 0.2 & 0.3 & 0.5 & 1 \end{pmatrix} \end{matrix}$$

(1,2) (1,3) (1,4) (2,3) (2,4) (3,4)

Reflexive ✓

Symmetric ✓

But Not Transitive

Tolerance Relation ✓

Before saying Equivalence Relⁿ you need to be very careful & want to all values then only it is equivalence.

Q →

$$R = \begin{bmatrix} 1 & 0.8 & 0.4 & 0.5 & 0.8 \\ 0.8 & 1 & 0.4 & 0.5 & 0.9 \\ 0.4 & 0.4 & 1 & 0.4 & 0.4 \\ 0.5 & 0.5 & 0.4 & 1 & 0.5 \\ 0.8 & 0.9 & 0.4 & 0.5 & 1 \end{bmatrix}$$

✓ Equivalence

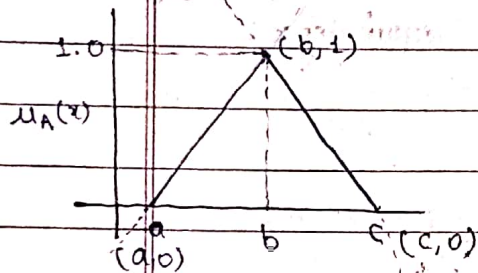
Membership Function

In fuzzy set theory, fuzzy set A of universe X is defined by function $\mu_A(x)$

* Triangular Membership Function

* Trapezoidal Membership Function

* Triangular MF :- Max value is 1 (Peak of triangular MF ≤ 1)



$$f_x(x) = \begin{cases} 0 & x \leq a \text{ or } x \geq c \\ \frac{x-a}{b-a} & a < x \leq b \\ \frac{(x-c)}{b-c} & b < x < c \end{cases}$$

$$y - 0 = \frac{1-0}{b-a} = m$$

$$y = \mu_A(x)$$

$$y = \frac{1}{b-a} (x-a)$$

$$\begin{aligned} (x_1, y_1) &\rightarrow (a, 0) & y - y_1 &= \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \\ (x_2, y_2) &\rightarrow (b, 1) \end{aligned}$$

$$y - 0 = \frac{1-0}{b-a} (x-a)$$

$$\text{Similarly, } y - 0 = \frac{1-0}{b-c} (x-b) = \frac{c-x}{c-b}$$

But how this graph is generated through coding

$$\mu(x) = \max \left(\min \left(\frac{x-a}{b-a}, \frac{c-x}{c-b} \right), 0 \right)$$

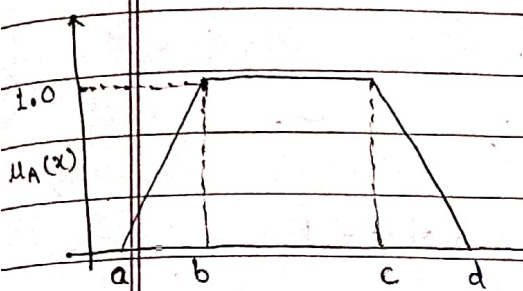
Here we need to ask a, b, c such that it is maximum on b i.e., 1.

we are building the ability of "linguistic behaviour" into machines with the help of fuzzy relation.

Try to find the utility of the things you are learning !!
Be & start thinking

* Trapezoidal Membership Function :

14/10/25



$$\mu_A(x) = \begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & b < x \leq c \\ \frac{x-d}{c-d} & c < x < d \\ 0 & x \geq d \end{cases}$$

$$\mu_A(x) = \max \left(\min \left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c} \right), 0 \right)$$

Gaussian Membership function :

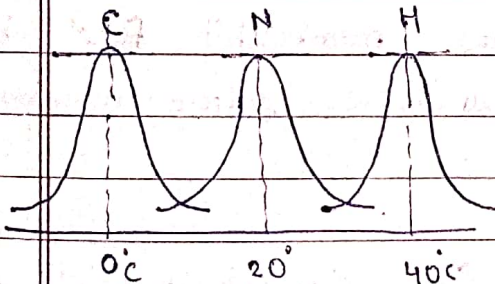
$$\mu_A(x) = e^{-\frac{(x-c)^2}{2\sigma^2}}$$

c = Mean σ = standard deviation

$\therefore f(x) = f(-x)$ Therefore it is even function.

The transition are very smooth.

Eg:



$$10^\circ \text{C} \Rightarrow (C, 0.3)$$

$$10^\circ \text{C} \Rightarrow (N, 0.3)$$

$$10^\circ \text{C} \Rightarrow (H, 0.01)$$

Generalised Bell Membership Function :

$$\mu_A(x) = \frac{1}{1 + \left(\frac{|x-c|}{a}\right)^{2b}}$$

a control the width
b control the slope
c is centre of bell curve

Sigmoid Membership Function :

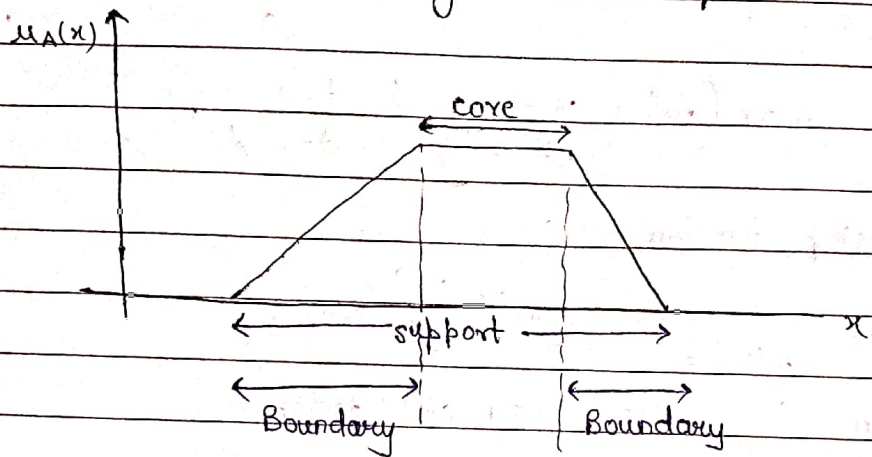
$$\mu_A(x) = \frac{1}{1 + e^{-a(x-c)}}$$

a → control slope of curve
c → centre of curve

Membership Function :

$$\mu_A(x) = 1$$

core : All member having membership value = 1

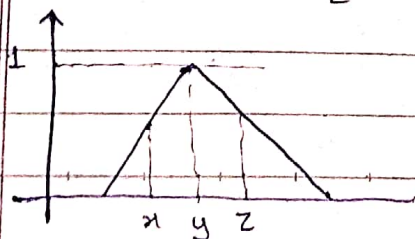


Support : Having $\mu_A(x) > 0$

Boundary : Having $0 < \mu_A(x) < 1$

Normal fuzzy set :- It is one whose membership funcⁿ has at least one element x in universe whose membership value is unity.

Convex set :- It is described for element x, y & z in fuzzy set A the relation $x < y < z$ implies that $\mu_A(y) \geq \min[\mu_A(x), \mu_A(z)]$



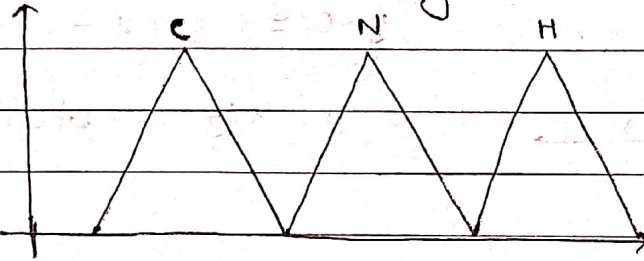
It satisfy for all point then it is convex fuzzy set.

Fuzzification :- In this the crisp input are converted into fuzzy set using membership function.

Page: _____
Date: _____

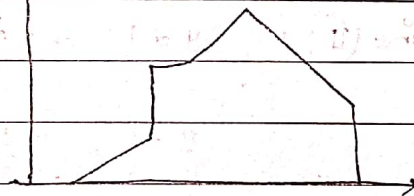
- Singleton fuzzification
- Gaussian "
- Trapezoidal or Triangular "
- Piecewise Linear "
- Sigmoidal "
- Max Min " (Threshold based fuzzification)
- Discretization

Singleton : The crisp input is associated with a specific degree of membership in only one fuzzy set.



Trapezoid : Triangular / trapezoidal membership function are used

Piecewise Linear :



Max Min / Threshold based fuzzification :- Rule based approach where membership is assigned by predefined threshold.

Defuzzification :- It refers to the conversion of fuzzy to crisp value.

Method :

- Lambda cut Method
- Weight average method
- Maxima "
- Centroid "

Lambda-cut - Method : Technique to convert fuzzy into crisp
 known as alpha cut method.

Page: _____
 Date: _____

$$A_\lambda = \{x \mid \mu_A(x) \geq \lambda\}$$

Eg: $A = \{(x_1, 0.9), (x_2, 0.5), (x_3, 0.2), (x_4, 0.3)\}$

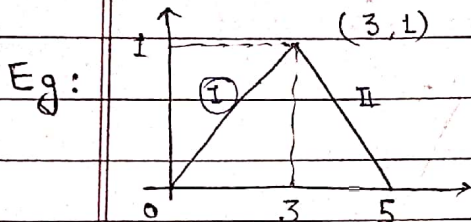
Then

$$A_{0.6} = \{(x_1, 1), (x_2, 0), (x_3, 0), (x_4, 0)\} = \{x_1\}$$

$$A_{0.4} = \{(x_1, 1), (x_2, 1), (x_3, 0), (x_4, 0)\} = \{x_1, x_2\}$$

$$A = \{(x_1, 0.1), (x_2, 0.5), (x_3, 0.8), (x_4, 0.7)\}$$

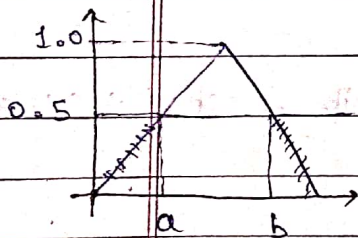
$$A_{0.2} = \{(x_1, 0), (x_2, 1), (x_3, 1), (x_4, 1)\} = \{x_2, x_3, x_4\}$$



$$y - 0 = \frac{1}{3}(x - 0)$$

$$y = x/3 \text{ --- Line (I)}$$

i) $\lambda = 0.5$



$$\because y = x/3 \text{ If } y = 0.5 = \lambda$$

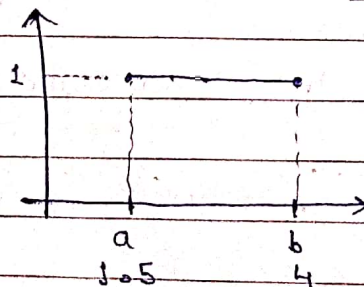
$$\therefore x = 1.5 = a$$

$$\text{Line (II): } y - 1 = \frac{0 - 1}{5 - 3}(x - 3)$$

$$y - 1 = -\frac{1}{2}(x - 3)$$

$$2y - 2 = -x + 3 \Rightarrow x + 2y = 5 \text{ --- Line (II)}$$

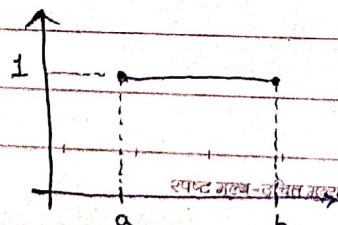
$$[x = 5 - 2y = 5 - 2 \times \frac{1}{2} = 5 - 1 = 4] \Rightarrow b = 4$$



ii) $\lambda = 0.6$

$$a = x = 1.8$$

$$b = x = 3.8$$



Eg:

$$\lambda = 0.6$$

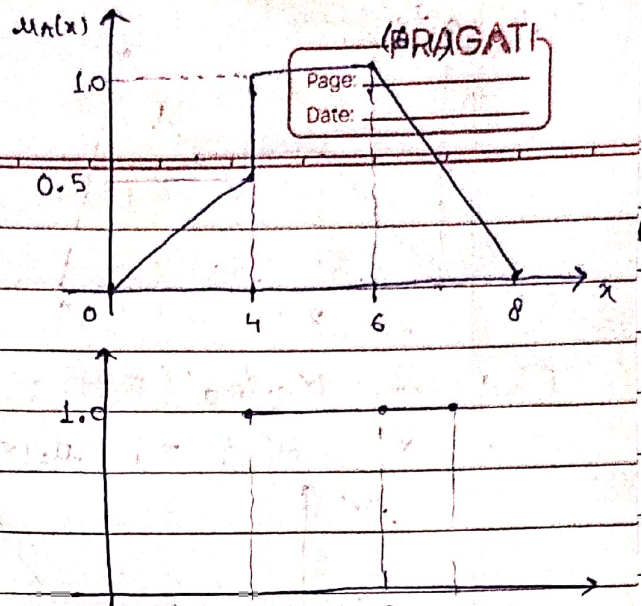
$$y - 1 = \frac{0 - 1}{8 - 6} (x - 6)$$

$$y - 1 = -\frac{1}{2} (x - 6)$$

$$2y - 2 = -x + 6$$

$$x + 2y = 8$$

$$x = 8 - 1.2 = 6.8$$



Q-7

$\mu_A(x)$	x_1	x_2	x_3	x_4	x_5
P	0.1	0.2	0.7	0.5	0.4
Θ	0.9	0.6	0.3	0.2	0.8

a) $P \cup \Theta$

$$P_{0.2} = \{x_2, x_3, x_4, x_5\}$$

$$\Theta_{0.3} = \{x_1, x_2, x_3, x_5\}$$

b) $P \cup \Theta = \{(x_1, 0.9), (x_2, 0.6), (x_3, 0.7), (x_4, 0.5), (x_5, 0.8)\}$

$$(P \cup \Theta)_{0.6} = \{x_1, x_2, x_3, x_5\}$$

c) $P \cup \bar{P} = \{(x_1, 0.9), (x_2, 0.8), (x_3, 0.7), (x_4, 0.5), (x_5, 0.4)\}$

$$(P \cup \bar{P})_{0.8} = \{x_1, x_2\}$$

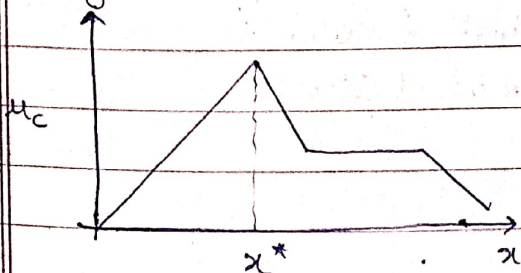
d) $P \cap \Theta = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3), (x_4, 0.2), (x_5, 0.4)\}$

$$(P \cap \Theta)_{0.4} = \{x_5\}$$

Maxima Method :

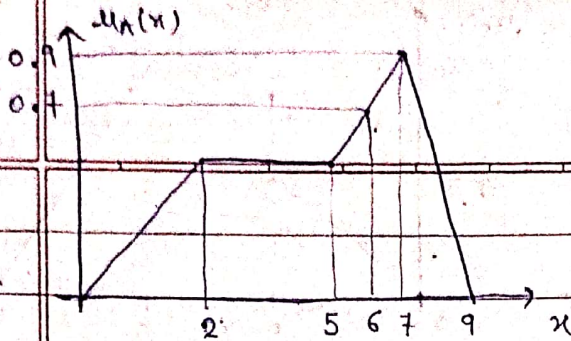
• Height method

$$\mu_c(x^*) \geq \mu_c(x) \text{ for all } x \in X$$



Applicable when height is unique.

Eg: $\mu_A(x)$



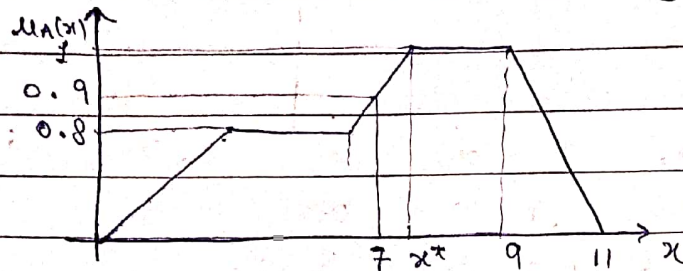
$$y - 0.4 = \left(\frac{0.7 - 0.4}{6 - 5} \right) (x - 5)$$

$$y - 0.4 = \frac{0.3}{1} (x - 5)$$

$$0.9 - 0.4$$

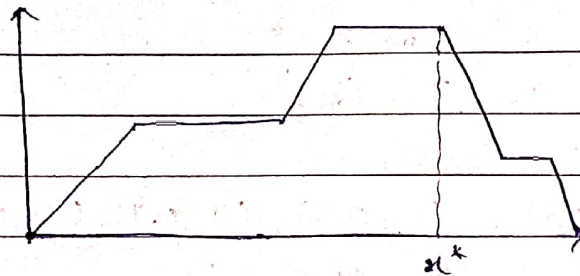
First of Maxima (FoM) :-

$$x^* = \min \{ x \mid \mu_c(x) = \max \text{ height of } \mu(x) \}$$

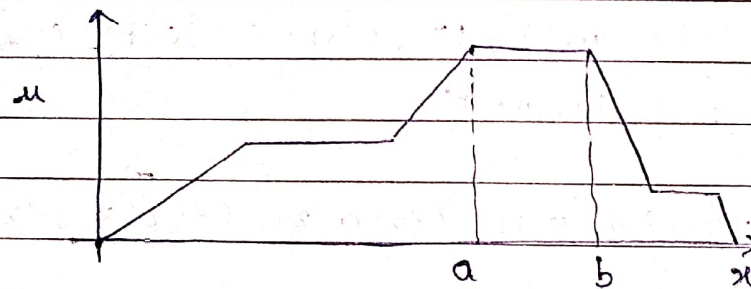


$$x^* = 8$$

Last of Maxima :



Mean of Maxima :-



$$x^* = \frac{a+b}{2}$$

Centroid Method

- Centre of gravity method (COG)
- Centre of sum method (COS)
- Centre of area method (COA)

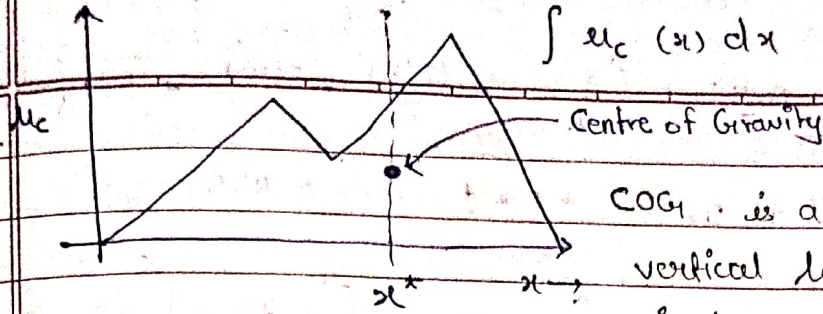
Centroid Method :

$$x^* = \frac{\int x \cdot u_c(x) dx}{\int u_c(x) dx}$$

Centre of gravity method

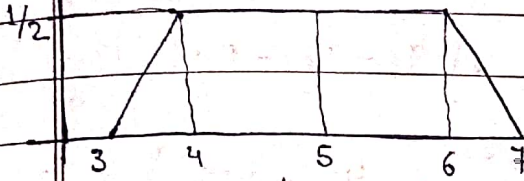
(Page:)

Date:



COG is a part of where vertical lines divide aggregate in two equal part.

Eg:

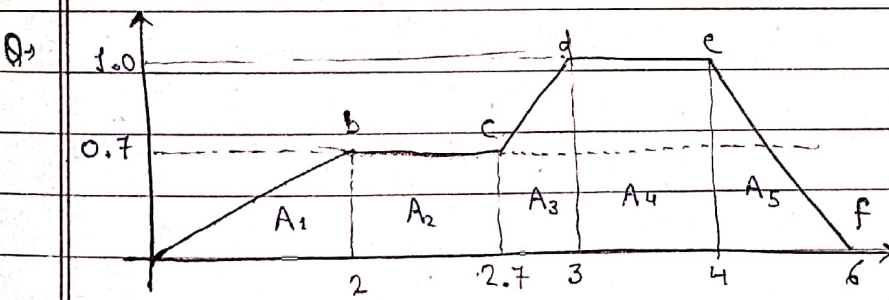


$$x^* = \frac{\int_3^4 x \cdot \left(\frac{1}{2}(x-3)\right) dx + \int_4^6 x \cdot \frac{1}{2} dx + \int_6^7 x \cdot \left(-\frac{1}{2}(x-7)\right) dx}{\int_3^4 \frac{1}{2}(x-3) dx + \int_4^6 \frac{1}{2} dx + \int_6^7 -\frac{1}{2}(x-7) dx}$$

$$x^* = \frac{\int_3^4 \frac{x^2}{2} - \frac{3x}{2} dx + \int_4^6 \frac{x}{2} dx - \frac{1}{2} \int_6^7 x^2 - 7x dx}{\int_3^4 \frac{x-3}{2} dx + \left(\frac{x}{2}\right)_4^6 + \int_6^7 \frac{7-x}{2} dx}$$

$$x^* = \frac{\frac{1}{2} \left(\frac{x^3}{3} - \frac{3x^2}{2} \right)_3^4 + \frac{1}{2} \left(\frac{x^2}{2} \right)_4^6 - \frac{1}{2} \left(\frac{x^3}{3} - 7x^2 \right)_6^7}{\frac{1}{2} \left(\frac{x^2}{2} - 3x \right)_3^4 + \frac{1}{2} \cdot 1 \cdot 2 + \frac{1}{2} \left(7x - \frac{x^2}{2} \right)_6^7}$$

$$x^* = 5$$



$$u_x = \begin{cases} \frac{0.7}{2}x & 0 < x \leq 2 \\ 0.7 & 2 < x \leq 2.7 \\ x-2 & 2.7 < x \leq 3 \\ 1 & 3 < x \leq 4 \\ -\frac{x}{2} + 3 & 4 \leq x \leq 6 \end{cases}$$

$$\int u_x(dx) = \int_0^2 \frac{0.7}{2}x dx + \int_2^{2.7} 0.7 dx + \int_{2.7}^3 x-2 dx + \int_3^4 1 dx + \int_4^6 -\frac{x}{2} + 3 dx$$

रूपरेखा ग्राह्य-वर्तमान ग्राह्य

Page: 4
Date: / /

$$= \frac{0.7}{2} \left(\frac{x^2}{2} \right)_0^{2.7} + 0.7(0.7) \cdot \left(\frac{x^2 - 2x}{2} \right)_{2.7}^3 + (x) \left(\frac{x^2}{2} \right)_3^{4.7} + \left(\frac{x^2}{2} \right)_4^{6.7} + \left(\frac{x^2}{2} \right)_5^{8.7}$$

$$= \frac{0.7}{2} \cdot \frac{4}{2} + 0.49 + \left(\frac{9}{2} - 6 - \frac{(2.7)^2}{2} + 5.4 \right) + 1 + \left(\frac{36}{2} - \frac{36}{2} \right)$$

$$= 0.7 + 0.49 + 0.255 + 1 + 1 = 3.445$$

$$\int_0^2 x + \frac{0.7}{2} x dx + \int_2^{2.7} 0.7x dx + \int_{2.7}^3 x^2 - 2x + \int_3^4 x dx + \int_4^6 -\frac{x^2}{2} + 3x dx$$

$$= \frac{0.7}{2} \left(\frac{x^3}{3} \right)_0^2 + \frac{0.7}{2} \cdot \left(\frac{x^2}{2} \right)_2^{2.7} + \left(\frac{x^3}{3} - x^2 \right)_{2.7}^3 + \left(\frac{x^2}{2} \right)_3^4 + \left(-\frac{x^3}{6} + \frac{3x^2}{2} \right)_4^6$$

$$= \frac{0.7}{2} \cdot \frac{1}{3} (8-0) + \frac{0.7}{2} \cdot \left(\frac{(2.7)^2}{2} - 4 \right) + \left((9-9) - \frac{(2.7)^3}{3} + \frac{(2.7)^2}{2} \right)$$

$$+ \frac{1}{2} (16-9) + \left(\frac{-6^3}{6} + \frac{3 \cdot 6^2}{2} + \frac{4^3}{6} - \frac{3 \cdot 4^2}{6} \right)$$

$$= 10.969$$

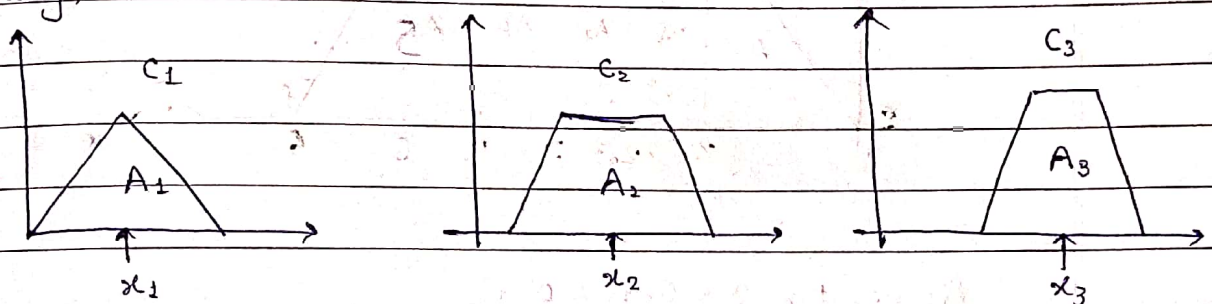
$$x^* = \frac{10.969}{3.445} = 3.18$$

Centroid Method : Center of Sum (Cos)

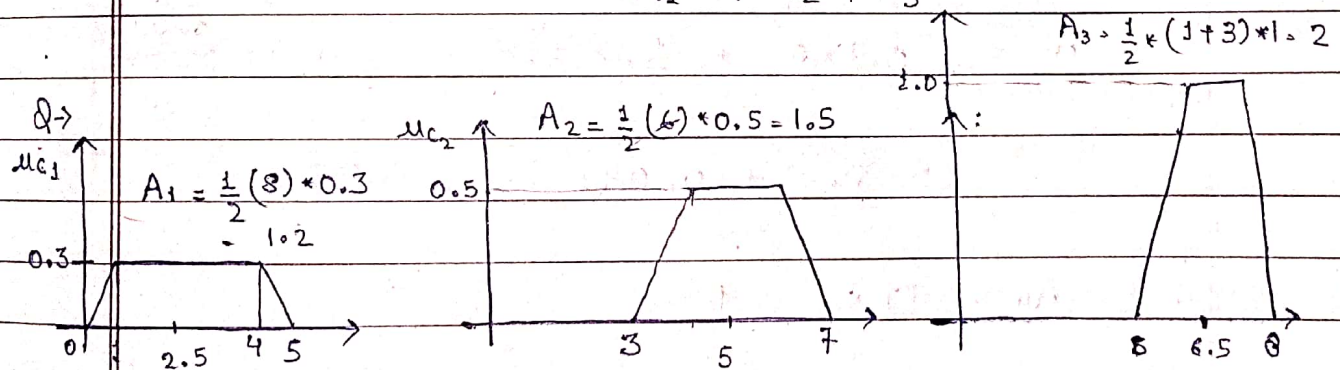
If the OIF fuzzy set $C = C_1 \cup C_2 \cup C_3 \dots C_n$ then the crisp value according to Cos is defined as

"Although the formula $x^* = \frac{\sum_{i=1}^n x_i A_i}{\sum_{i=1}^n A_i}$ is discrete but yet applied on continuous graph" $A_i \rightarrow$ area of the region bounded by the fuzzy set C_i
 $x_i \rightarrow$ Geometric center of the area A_i

Graphically,



$$x^* = \frac{x_1 A_1 + x_2 A_2 + x_3 A_3}{A_1 + A_2 + A_3}$$



$$x^* = \frac{0.3 \times 2.5 + 0.5 \times 5 + 1 \times 6.5}{2.5 + 5 + 6.5} = \frac{0.75 + 2.5 + 6.5}{14}$$

$$x^* = 2.5 \times 1.2 +$$

For C_1 ,

$$x_1 = \frac{0+5}{2} = 2.5$$

$$A_1 = \frac{1}{2} (a+b)h = \int_0^5 \mu_{C_1}(x) dx$$

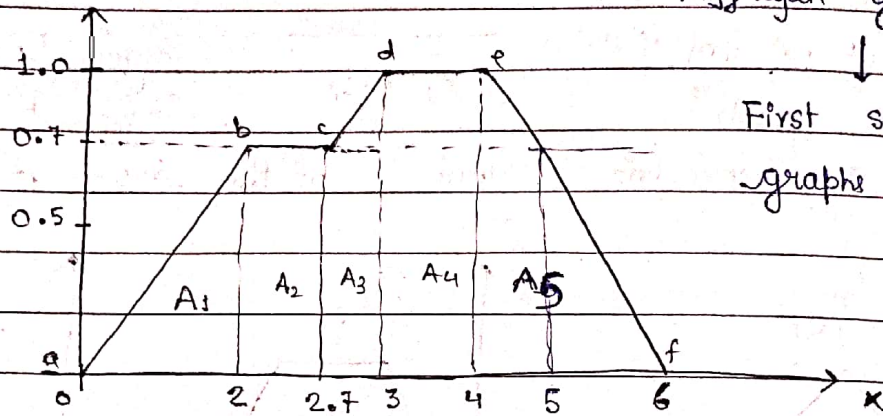
$$x^* = \frac{2.5 \times 1.2 + 5 \times 1.5 + 6.5 \times 2}{1.2 + 1.5 + 2}$$

Page: _____
Date: _____

PRAGATI

$$= \frac{7.08 + 7.5 + 13}{4.7} = \frac{23.5}{4.7} = 5$$

Q. Find out x^* for this



Aggregate graph is given

First separate out all graphs

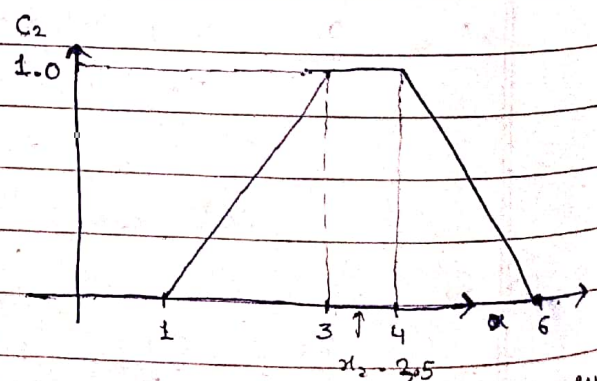
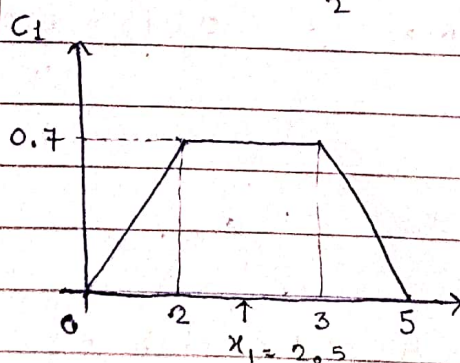
$$A_1 = \frac{1}{2} \times 2 \times 0.7 = 0.7$$

$$A_2 = 0.7 \times 0.7 = 0.49$$

$$A_3 = 0.7 \times 0.3 + \frac{1}{2} \times 0.3 \times 0.3 = 0.21 + \frac{0.09}{2} = 0.21 + 0.045 = 0.255$$

$$A_4 = 1 \times 1 = 1$$

$$A_5 = \frac{1}{2} \times 2 \times 1 = 1$$



These figures are not unique $\rightarrow$ but their union will give the aggregate

COG

COS

COA

WA

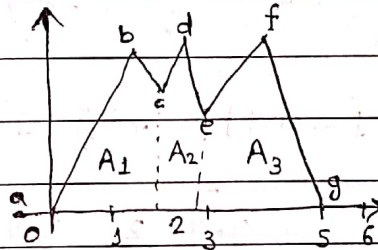
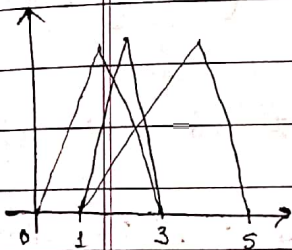
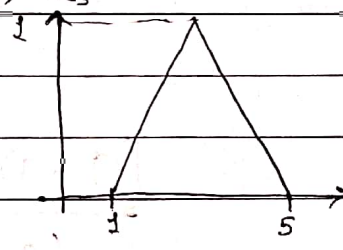
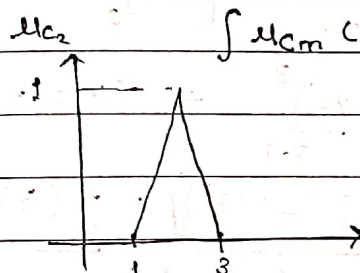
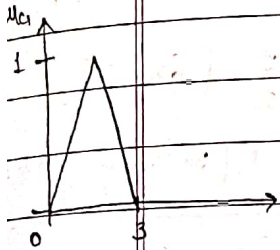
For a particular question we need to compare all these things.

COS (Centre of Sum): geometric centre of area A_c .

$$x^* = \frac{\sum_{i=1}^n x_i A_{c_i}}{\sum_{i=1}^n A_{c_i}}$$
 Area bounded by fuzzy set c_i

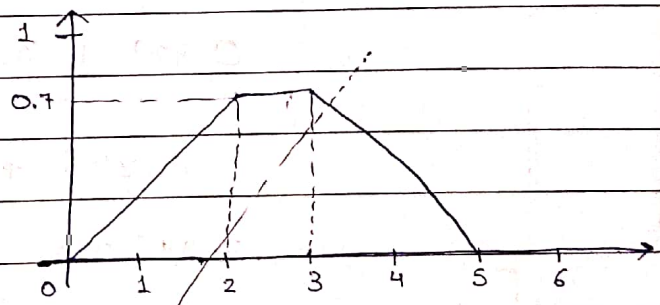
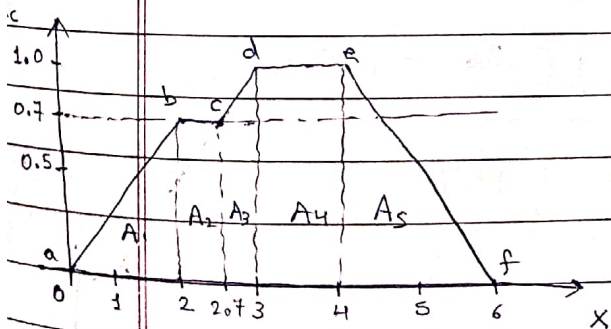
Centroid Methods: CoA (Centre of Largest area)

$$x^* = \frac{\int x \mu_{cm}(x) dx}{\int \mu_{cm}(x) dx}$$



Apply CoA on this graph by knowing the line eqⁿ from PPT

You need to start from aggregate → then you need to calculate the area
 Now, we need to obtain the largest area



How area segments is find? → first segment will be $A_1 + A_2$

Because from initial values it is increasing $a \rightarrow b$ & $b \rightarrow c$ constant (it is not increasing) so it is considered it a single segment.

If values starts increasing from that point will start

∴ 2 ^{area} segments will be there.

COG of largest area only

$$x^* = \frac{\int_0^6 x \cdot u_{cm}(x) dx}{\int_0^6 u_{cm}(x) dx}$$

(2, 0.7) (3, 1)

$$y - 0.7 = \frac{1 - 0.7}{3 - 2.7} (x - 2.7)$$

$$y - 0.7 = \frac{0.3}{0.3} (x - 2.7) \quad x^* = 2.7$$

$$y = x - 2.0$$

(4, 1), (6, 0)

$$y - 1 = \frac{0 - 1}{6 - 4} (x - 4)$$

$$2y - 2 = -x + 4$$

$$2y = -x + 6$$

$$y = \frac{6 - x}{2}$$

$$= \int_{2.7}^3 x(x-2) dx + \int_3^4 x \cdot 1 dx + \int_4^6 x \cdot \frac{(6-x)}{2} dx$$

$$= \int_{2.7}^3 (x^2 - 2x) dx + \int_3^4 x dx + \int_4^6 \frac{6x - x^2}{2} dx$$

$$= \left[\frac{x^3}{3} - x^2 \right]_{2.7}^3 + \left[\frac{x^2}{2} \right]_3^4 + \frac{1}{2} \left[3x^2 - \frac{x^3}{3} \right]_4^6$$

$$= \left(\frac{27}{3} - 9 \right) - \left(\frac{2.7^3}{3} - 2.7^2 \right) + \left(\frac{16}{2} - \frac{9}{2} \right) + \frac{1}{2} \left[\frac{3 \cdot 36 - 216}{2} - \frac{3 \cdot 16 - 64}{2} \right]$$

$$= \left(\frac{9}{2} - 6 - \frac{2.7^2}{2} + 5.4 \right) + 1 + \frac{1}{2} \left[\frac{36 - 36 - 24 + 16}{2} \right]$$

$$= \frac{-12 + 5.4}{2} + \frac{2.7}{2} + \frac{1}{2} \left[\frac{36 - 36 - 48 + 64}{2} \right]$$

$$= \frac{-3}{2} + 2.7 \left(\frac{2 - 2.7}{2} \right) + 1 + \frac{1}{2} \times \left(\frac{36 - 16}{2} \right)$$

$$= \frac{-3}{2} + 2.7 \times \frac{1.3}{2} + 1 + 1$$

$$= 0.729 + 3.5 + \frac{1}{2} [36 - (26.66)]$$

$$0.225 + 1 + \frac{1}{2} [2]$$

$$3.998052$$

Steps badaho → Accuracy Badhegi!!

— DR. SHARMA

$$\left[\frac{3^3}{3} - 3^2 - \left(\frac{2 \cdot 7^3}{3} - 2 \cdot 7^2 \right) \right] + \frac{1}{2} [16 - 9] + \frac{1}{2} \left[3 \cdot 6^2 - 6^3 - \left(3 \cdot 4^2 - 4^3 \right) \right]$$

$$\left(\frac{3^3}{2} - 2 \cdot 3 \right) - \left(\frac{2 \cdot 7^2}{2} - 2 \cdot 2 \cdot 7 \right) + 1 + 1$$

$$0.729 + 3.5 +$$

$$\approx 3.998052$$

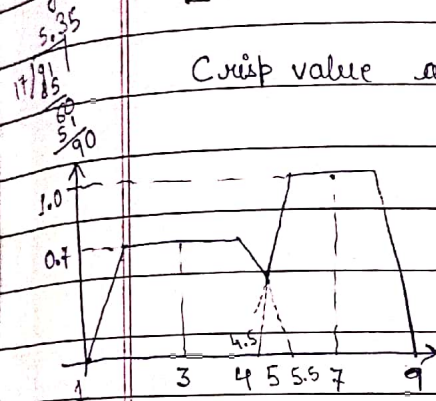
DR. SUDEEP SHARMA

03/11/25

Weighted average method $\rightarrow$ used for symmetrical o/p membership function.
Applicable on all individual elements.

Crisp value according to this method is $\Rightarrow x^* = \frac{\sum_{i=1}^n \mu_{C_i}(x_i) \cdot (x_i)}{\sum_{i=1}^n \mu_{C_i}(x_i)}$

$x_i \rightarrow$ middle of the fuzzy set C_i

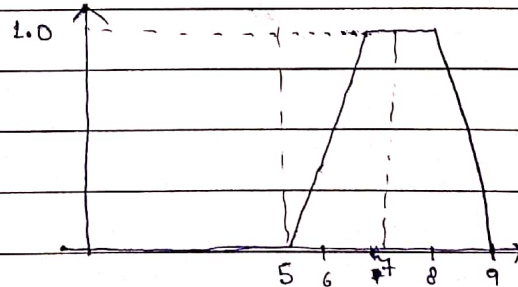
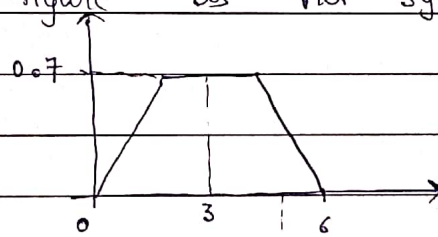
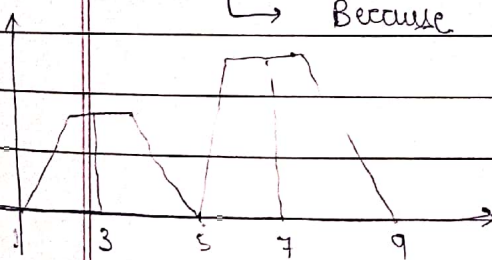


$$x^* = \frac{0.7 \times 3 + 1 \times 7}{0.7 + 1.0} = \frac{9.1}{1.7}$$

$$x^* = 5.35$$

$\therefore$ Ideally, for this type of diagram we do not apply weighted average & sum method.

$\rightarrow$ Because the figure is not symmetrical

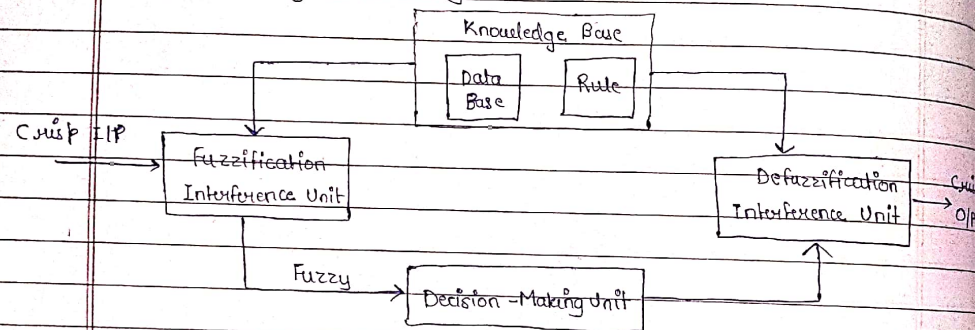
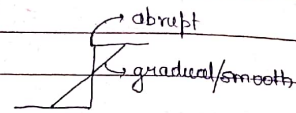


Fuzzy Inference systemFuzzy Decision Making System

Q → What is the difference b/w fuzzy decision making system & traditional decision making system?

Fuzzy Inference system (FIS):

- Smooth transition
- Human like reasoning
- Adaptability / Flexibility

FIS

Mamdani FIS

Sugeno FIS

→ The Sugeno FIS, defuzzification Inference unit is absent / avoided

TIPPING

- * First we need to think
- * Identify the inputs
- * poor → MF → service
- good
- excellent

For output, Tip

- * Universe of discourse consider from your own

- * import useful libraries

- * Then make user defi

End Sem

You need to plot

Fuzzy Rule Base

Low

Low Exzellen

Temperature Normal Exzellen

High Normal

O/P

TIPPING PROBLEM

* First we need to think about I/P, O/P, Universe of discourse

* Identify the inputs $\rightarrow$ Food quality
 $\rightarrow$ service quality

* poor $\rightarrow$ MF $\rightarrow$ service
 good $\nearrow$
 excellent $\nearrow$

varied $\rightarrow$ MF $\rightarrow$ Food
 delicious $\nearrow$

For output, Tip $\rightarrow$ MF's are $\rightarrow$ cheap
 $\rightarrow$ average
 $\rightarrow$ generous

* Universe of discourse will be given in question or you have to consider from your own.

DR. SUDEEP SHARMA

11/11/25

* import useful libraries $\rightarrow$ numpy
 $\rightarrow$ matplotlib

* Then make user defined function for triangular, trapezoidal & etc.

End Sem

You need to plot & represent in exam

Fuzzy Rule Base (Comfort)

Humidity

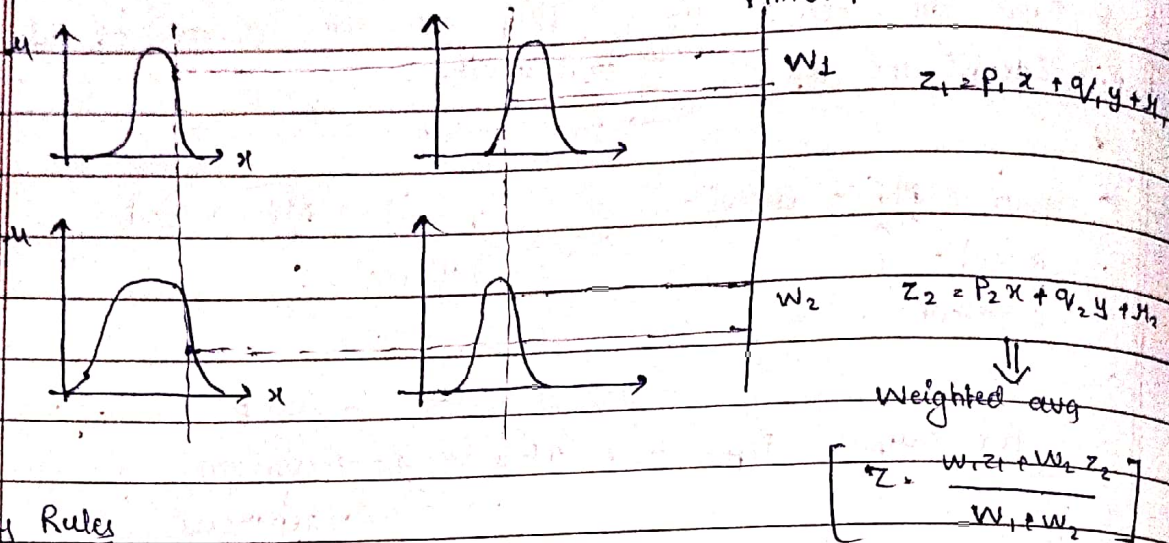
	Low	Normal	High
Low	Excellent	Normal	Poor
Normal	Excellent	Normal	Poor
High	Normal	Poor	Poor

[Quiz $\rightarrow$ Wednesday]

O/P : Comfort.

Sugeno Fuzzy Inference system :-

→ Here defuzzification step is not required.

Fuzzy Rules

In a Sug.

Example: Let $x = 1.5$, $y = 2$.

Rule 1: If x is low and y is high, $z_1 = 2x + y + 1$

Rule 2: If x is high & y is low $z_2 = x - 3y + 2$

$$\mu_{A1}(x) = 0.6 \quad \mu_{B1}(y) = 0.8$$

$$\mu_{A2}(x) = 0.4 \quad \mu_{B2}(y) = 0.5$$

$$w_1 = \mu_{A1}(x) * \mu_{B1}(y) = 0.6 * 0.8 = 0.48$$

$$w_2 = \mu_{A2}(x) * \mu_{B2}(y) = 0.4 * 0.5 = 0.2$$

$$z = 0.48(2x + y + 1) + 0.2(x - 3y + 2)$$

$$0.68$$

$$= 48(2x + y + 1) + 20(x - 3y + 2)$$

$$68$$

$$= 3 * 48 + 20 * (-2.5)$$

$$68$$

$$34$$

$$\left[z = \frac{144 - 25}{34} = \frac{119}{34} = 3.5 \right]$$

PID $\rightarrow$ Proportional Integral Development Differential

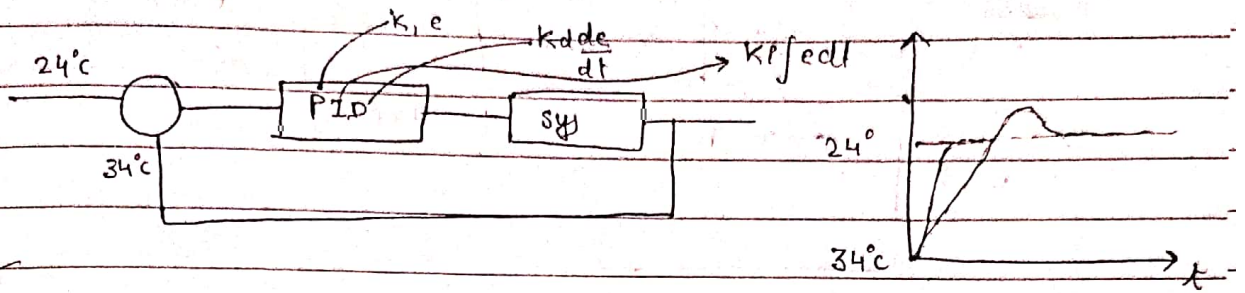


Diagram Fuzzy Control system

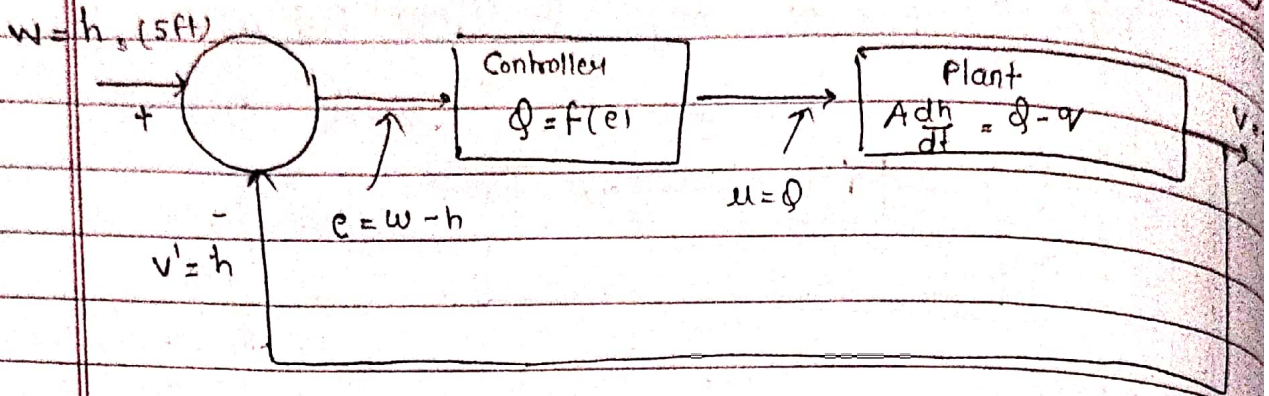


Fig: Closed Loop liquid control system

- # Calculating the time delay in the system is very tedious task
- # Controller doesn't stop even if error is zero.

↳ Because making the error zero is the objective of the controller.

- # Use CHATGPT intelligently to become more knowledgeable otherwise you can't imagine the consequences....

Development of Membership Functions :-

Intuition → based on normal human behaviour

Inference → Intuition + Fact/Knowledge/Reasoning

Q → In identification of a triangle, A, B, C inner angles
 $U = \{ (A, B, C) \mid A \geq B \geq C \geq 0 ; A + B + C = 180^\circ \}$

For fuzzy right angle triangle, the membership value is as :

$$\frac{0.94}{1.130} = \frac{162}{80}$$

$$\frac{10}{90} = \frac{1}{9}$$

$$\mu_{(A,B,C)} = 1 - (10) - (-10)$$

$$\mu_{(A,B,C)} = 1 - (A - 90) - (B + C - 90)$$

$$= 1 - |A - 90|$$

for some values of A it is giving correct but for all angles it is not

$$= 1 - \frac{|A - 90|}{90}$$

So we have to analysis different possibilities & calculate which one is much more accurate.

Q.1

Try Formulation for equilateral triangle &
& also for square & rectangle & circle.

isoscPRAGATI
Page: _____
Date: _____